

Study of Ecosystems**Aim**

To study different ecosystems in the surrounding for a comparative understanding.

Requirements

Writing material, binoculars, a shovel, troughs, beakers or any other containers for collection.

Procedure

1. Make sure you understand the concept of an ecosystem before choosing one to study.
2. Choose an ecosystem to study – this could be a huge tree, a lake, a pond, a certain area of a field, your own school playground, a butterfly garden (if present on your school campus), a big aquarium, a coastal strip, a beach, etc.
3. Never disturb any natural ecosystem by plucking plant parts or harming any animals or insects. Take good care of all the resident fauna and flora while carrying out your study. You may photograph the specimens, or collect fallen leaves, flowers, or seeds instead.
4. Collect soil samples using a shovel so that you can analyse the pH, colour, etc. of the soil at the school laboratory. Enter these details into the tabulations given below.
5. Also think about conservation measures to protect these environments – natural ecosystems today are being destroyed in the pursuit of development. You may focus part of your work on some conservation measures.

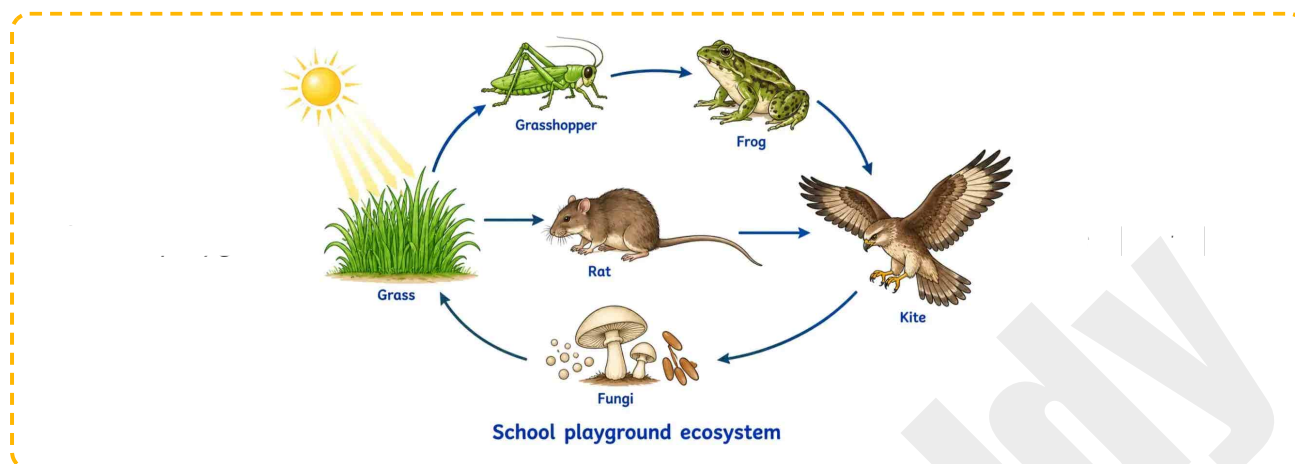
Observations

Ecosystem	Ecosystem – 1	Ecosystem – 2
Date or Period of study	Rainy season	Rainy season
Name of Ecosystem	School play ground	Pond
Abiotic factors	Soil, water, sunlight, air	Water, sunlight, mud, stone
Biotic factors (Flora and Fauna)	Plants, grasshopper, frog, earthworm	Algae, frog, snake, aquatic plants

General description	Ecosystem of school playground is healthy.	Ecosystem in pond is also found healthy.
Conclusion	Conservation and plantation is required.	We can protect pond ecosystem by avoiding contamination.

Diagram

(Draw a food web or food chain from the ecosystem that you have studied)



Conclusion

(Draw the inference about the health of the ecosystem that you have studied. Check up whether this ecosystem is sustaining or is threatened due to human activities.)

Out of the ecosystems that I have studied, I found pond ecosystem as the most healthy and balanced and in good condition, because, Pond water was clean and without any garbage. People were taking care of pond ecosystem. Notice boards were displayed mentioning about not to throw garbage in pond and surrounding area. Polluting pond water by washing utensils, clothes, bathing, etc. was prohibited.

Balanced proportion of biotic as well as abiotic factors was observed in pond ecosystem.

Multiple Choice Questions

1. The aquatic ecosystems are worst affected by which of the following human interference?

- (A) Noise pollution (B) Plastic Pollution
(C) Deforestation (D) Construction activities

Answer: (B) Plastic Pollution

2. Which abiotic factor is the most important for the survival of animals?

- (A) Oxygen content (B) Nitrogen in air
(C) Carbon dioxide concentration (D) Soil structure

Answer: (A) Oxygen content

3. Which of the following are autotrophic organisms found in sea?

(A) Whale (B) Phytoplanktons (C) Fishes (D) Turtles

Answer: (B) Phytoplanktons

Remember

An 'ecosystem' is a region with specific and recognizable landscape form such as forest, grassland, desert, wetland or coastal area.

Date: _____

Teacher's Signature: _____